Introduction to Artificial Intelligence

Module 1 — AI Engineer Bootcamp

1. What is Artificial Intelligence?

Artificial Intelligence (AI) refers to the simulation of human intelligence in machines that are programmed to think and learn. The term was coined by John McCarthy in 1956 during the Dartmouth Conference, which is widely considered the birthplace of AI as a field.

AI systems can perform tasks that typically require human intelligence, such as understanding natural language, recognising patterns, solving problems, and making decisions. AI is broadly categorised into two types: Narrow AI (designed for a specific task) and General AI (capable of performing any intellectual task a human can do).

2. A Brief History of AI

The history of AI spans several decades of progress, setbacks, and breakthroughs:

Alan Turing, often called the father of computer science, proposed the concept of a machine that could simulate human reasoning in his 1950 paper 'Computing Machinery and Intelligence'. He introduced the famous Turing Test as a measure of machine intelligence — if a machine could converse indistinguishably from a human, it could be considered intelligent.

In the 1950s and 1960s, early AI programs like the Logic Theorist and General Problem Solver demonstrated that machines could perform symbolic reasoning. The 1970s and 1980s saw the rise of expert systems — programs that encoded human expertise as rules.

The field experienced periods known as AI winters, where funding and interest declined due to unmet expectations. However, the 2000s and 2010s brought an AI renaissance driven by big data, powerful GPUs, and advances in machine learning.

3. Key Concepts in AI

Machine Learning (ML) is a subset of AI that enables systems to learn from data without being explicitly programmed. Instead of writing rules, developers feed data to algorithms that discover patterns automatically.

Deep Learning is a subset of machine learning that uses artificial neural networks with many layers (hence 'deep'). These networks are inspired by the human brain and are particularly powerful for tasks like image recognition and natural language processing.

Natural Language Processing (NLP) is the branch of AI that enables computers to understand, interpret, and generate human language. It powers applications like chatbots, translation services, and voice assistants.

Computer Vision allows machines to interpret and understand visual information from images and videos. Applications include facial recognition, medical imaging, and self-driving vehicles.

4. Neural Networks

A neural network is a computational model loosely inspired by the structure of the human brain. It consists of layers of interconnected nodes (neurons). The input layer receives data, hidden layers process it, and the output layer produces the result.

Training a neural network involves feeding it labelled data and adjusting the weights of connections through a process called backpropagation. The network learns to minimise the difference between its predictions and the actual answers using an optimisation algorithm like gradient descent.

The more data and computing power available, the more complex and accurate neural networks can become. This has led to the development of large language models (LLMs) like GPT, which can generate human-like text.

5. Large Language Models (LLMs)

Large Language Models are a type of deep learning model trained on massive amounts of text data. They learn statistical patterns in language and can generate coherent, contextually relevant text.

GPT (Generative Pre-trained Transformer), developed by OpenAI, is one of the most well-known LLMs. It uses the Transformer architecture, which relies on an attention mechanism to understand relationships between words in a sentence regardless of their distance.

LLMs have applications in chatbots, code generation, content creation, summarisation, question answering, and much more. They represent a major milestone in AI development.

6. The Transformer Architecture

The Transformer model, introduced in the 2017 paper 'Attention Is All You Need' by Vaswani et al., revolutionised natural language processing. Unlike previous recurrent models, Transformers process entire sequences simultaneously using a mechanism called self-attention.

Self-attention allows the model to weigh the importance of each word in a sentence relative to all other words. This enables the model to capture long-range dependencies and contextual meaning far more effectively than earlier approaches.

Today, virtually all state-of-the-art language models, including BERT, GPT, and T5, are based on the Transformer architecture.

7. AI Ethics and Responsible AI

As AI becomes more powerful and widespread, ethical considerations are increasingly important. Key concerns include bias in AI systems, where models trained on biased data can perpetuate or amplify unfair outcomes.

Privacy is another critical issue, as AI systems often require large amounts of personal data. Transparency and explainability — the ability to understand why an AI made a particular decision — are essential for building trust.

Responsible AI development involves ensuring systems are fair, accountable, transparent, and do no harm. Organisations like Anthropic, OpenAI, and DeepMind invest in AI safety research to ensure AI remains beneficial to humanity.

8. Applications of AI

AI is transforming nearly every industry. In healthcare, AI assists with disease diagnosis, drug discovery, and personalised medicine. In finance, it powers fraud detection, algorithmic trading, and credit scoring.

In transportation, AI enables autonomous vehicles and optimises logistics. In education, adaptive learning systems personalise content for individual students. In entertainment, AI generates music, art, and video content.

The widespread adoption of AI raises important questions about the future of work, as many routine tasks become automated. However, AI also creates new roles and opportunities, particularly in AI engineering, data science, and machine learning operations.

9. Getting Started with AI Engineering

Becoming an AI engineer requires a combination of programming skills, mathematical foundations, and knowledge of AI frameworks. Python is the dominant language in AI due to its simplicity and rich ecosystem of libraries.

Key libraries include NumPy and Pandas for data manipulation, Matplotlib for visualisation, Scikit-learn for classical machine learning, PyTorch and TensorFlow for deep learning, and LangChain for building LLM-powered applications.

A strong understanding of linear algebra, calculus, probability, and statistics is essential. These mathematical concepts underpin how machine learning algorithms learn from data.

10. Summary

In this module, we explored the foundations of Artificial Intelligence — from its historical roots with Alan Turing to modern large language models and Transformer architectures. We covered key concepts including machine learning, deep learning, neural networks, NLP, and responsible AI.

This knowledge forms the foundation for everything that follows in the AI Engineer Bootcamp. As you progress through the course, you will gain hands-on experience building real AI-powered applications using Python, LangChain, vector databases, and more.